AI PLACEMENT OS

Universal Realtime Voice Agent

FINAL MASTER SPECIFICATION

PRODUCT DEFINITION

A realtime, user-scoped AI career operating layer that can understand natural speech, access the authenticated user's Placement OS data, access live external career intelligence, navigate and operate supported app capabilities, and verify every important action.

Scope

- Realtime voice conversation
- Complete personal Placement OS context
- Live jobs, companies, market and technology intelligence
- Universal in-app capabilities through controlled tools
- Memory, planning, safe actions and verification
- Security, privacy, reliability and adversarial testing

Status: Feature definition frozen for implementation planning

Deployment and publishing remain intentionally paused until the feature set is complete.

Prepared: 7 October 2026 | Basis: current AI Placement OS repositories, existing phase history, and current official platform documentation/research

01

Executive Summary

The complete product behavior and engineering contract in one place.

What the agent is

The Voice Agent is not a voice chatbot attached to one feature. It is the universal conversational interface to the AI Placement OS. A user can talk naturally about career goals, jobs, preparation, interviews, applications, skills and progress, and the agent can retrieve the right personal data, consult the live career world, reason across both, navigate the app, and execute supported actions safely.

Dimension	What the agent knows / does	Target behavior
Personal	The authenticated user's Placement OS data	Understands the user as an individual, not as a generic candidate.
World	Current jobs, companies, roles, skills, technologies, market and learning resources	Uses live retrieval for changeable facts instead of relying on model memory.
App control	Navigation and supported Placement OS actions	Can carry out app capabilities through a controlled tool layer.
Conversation	Natural realtime dialogue	Streaming speech, turn detection, interruption and recovery.
Safety	Authorization, ownership, confirmation, validation, provenance	The model is not the security boundary.
Reliability	Reconnect, cancel, resume, deduplicate, verify	No silent duplicate actions or false success claims.

QUALITY PROMISE

We cannot honestly promise mathematical zero-bug software. Our release bar is stronger and testable: no known P0/P1 defect, every critical invariant exercised, deliberate regressions caught by mutation/negative tests, critical realtime flows tested on real devices, and no claim of success without evidence.

Non-negotiable principles

- **Realtime first.** Conversation must feel live: streaming audio, low-latency turn taking, interruption and recovery are core architecture, not polish.
- **User-scoped by construction.** Identity comes from authenticated backend context; the model never chooses an arbitrary user/profile identifier.
- **Tools, not raw database access.** The agent requests capabilities; existing domain services enforce ownership and business rules.

- **World facts are fresh.** Jobs, hiring, market demand and technology trends are retrieved dynamically with source and freshness metadata.
- **Plan -> execute -> verify.** A tool call is not proof of success. Important actions are confirmed and their results verified.
- **Existing hardening is reused.** Idempotency, concurrency, AI boundaries, input/output guards, error classification and lifecycle recovery remain shared infrastructure.
- **Deployment is out of scope for now.** Signing, production backend deployment and publishing stay paused until feature work is finished.

02 - Product Map

A single agent exposes three connected knowledge planes: the user, the career world, and the app itself.

Plane	Purpose	Examples
Personal Placement OS	Know the authenticated user and their career state.	Profile, resume, ATS, skills, projects, GitHub, Career Twin, goal, path, strategy, jobs, applications, interviews, assessments, weaknesses, missions, progress.
World Career Intelligence	Know what is changing outside the app.	Live jobs, companies, role requirements, technology demand, hiring trends, market signals, salary signals where trustworthy, learning resources, official documentation.
App Capability Layer	Do what the app supports.	Open screens, save jobs, update applications, add notes, generate preparation, start interviews, create reminders, update supported settings.

The core loop

Step	Agent behavior
1. Hear	Receive streaming microphone audio; detect speech and turns.
2. Understand	Determine intent, entities, current-screen context and whether the user is asking for information or an action.
3. Retrieve	Call only the relevant personal and/or world tools.
4. Reason	Combine verified results with the conversation state and user goal.
5. Plan	For multi-step tasks, construct a bounded plan with dependencies.
6. Act	Execute approved capabilities using existing domain services and idempotency protections.
7. Verify	Confirm the authoritative result before claiming success.

Step	Agent behavior
8. Speak	Stream the answer back through TTS while keeping interruption possible.
9. Learn safely	Update approved memory or Placement OS state only through explicit, governed mechanisms.

USER EXPERIENCE TARGET

The user should be able to talk to the Placement OS as if talking to a highly capable career coach who already knows their profile and can look at the current job market.

03 - Complete Personal Placement OS Access

The agent can access all career-relevant data that the logged-in user is already authorized to access in the app.

Identity and profile

- Profile
- Education
- Experience
- Career preferences
- Career goals
- Target roles
- Preferred locations

Resume and ATS

- Uploaded resumes
- Parsed resume content
- ATS analyses
- ATS history
- Resume recommendations
- Resume weaknesses

Skills and evidence

- Skills
- Skill gaps
- Skill Passport
- Evidence
- Role-skill relationships
- Progress

Projects and professional presence

- Projects
- Project Builder
- Project proof/evidence
- GitHub connection

- GitHub repositories/evidence
- Professional Presence audit

Career intelligence

- Career Twin
- Career Path
- Career Strategy
- Career Decision Lab
- Bottlenecks
- Contradictions
- Next Best Action
- Readiness/evidence signals

Jobs and applications

- Live/discovered jobs
- Saved jobs
- Job matches
- Saved searches
- Job alerts
- Manual jobs
- Job preparation
- Application statuses
- Application notes
- Application outcomes

Interview and assessment intelligence

- Interview history
- Interview questions
- Answers
- Evaluations
- Full Recruiter Simulation
- Recruiter Attack
- Assessments
- Weaknesses

Execution and learning

- Roadmaps
- Preparation Plans
- Daily Missions
- Placement Path
- Placement Alarm
- Notifications
- Progress

SECURITY BOUNDARY

"Access to all user data" means all data the authenticated user is authorized to access through Placement OS. It does not mean unrestricted database access, secrets, other users, OAuth tokens, passwords, signing keys, provider credentials or arbitrary device data.

04 - World Career Intelligence

The agent also needs a live view of the external career world so it can answer current questions accurately.

Capability	Questions it should answer	Freshness requirement
Live job search	What jobs are available now? Which companies are hiring freshers? Find Hyderabad/UAE/remote roles.	Very fresh; query current sources.
Role intelligence	What does a Data Engineer do? What skills are expected? What is entry-level vs senior?	Refresh as role requirements change.
Company intelligence	Is Company X hiring? What roles and skills are they currently asking for?	Prefer official company sources or strong current providers.
Technology intelligence	What technologies are growing? Should I learn Spark, Kubernetes, GenAI or another stack?	Time-sensitive; show evidence date.
Market intelligence	Which roles/skills are growing in India, UAE or another market?	Geography-specific and dated.
Learning intelligence	What should I learn and which resources are best for this target role?	Prefer current official docs and reputable resources.
Interview intelligence	What topics commonly matter for this role/company?	Use current public information and label uncertainty.

World tools

- search_web and approved public sources
- search_jobs / provider-backed job retrieval
- search_company_jobs and company career data
- get_role_requirements and role knowledge
- get_skill_demand and technology trends
- get_location_market and market comparison
- search_learning_resources and official documentation
- search_public_interview_topics

Source metadata required

Field	Purpose
Source	Where the fact came from.
Retrieved at	When our system observed the result.
Published/updated at	When the source says it changed, if available.

Field	Purpose
Geography	India, UAE, US, etc. Prevents mixing regional evidence.
Provider	Which adapter or source supplied it.
Freshness state	Fresh, aging, stale, refresh required.
Confidence	Known, inferred, or insufficient evidence.

05 - Realtime Voice Experience

Realtime is a hard P0 requirement for both conversation and current-world retrieval.

Realtime capability	Required behavior
Streaming audio	Microphone input and spoken output are streamed; avoid the main-agent pattern of recording a whole file and waiting.
VAD / turn detection	Detect when the user starts and finishes speaking; tune for technical vocabulary and normal pauses.
Barge-in	User speech cancels the agent response immediately and preserves conversational state.
False interruption handling	Backchannels such as "okay" or "mm-hmm" should not constantly terminate responses.
Preemptive generation	Where safe, begin preparing responses near the end of a likely turn; cancel if the user continues.
Audio focus	Handle calls, alarms, media and other audio sources deliberately.
AEC / noise handling	Reduce echo and background noise before VAD/STT.
Device switching	Handle Bluetooth, wired headsets and speaker changes without losing the session.
Reconnect	Recover from transient network or transport failures.
Session resumption	Resume long conversations without resetting the user context.
Cancellation propagation	Stop audio, model work and safe-to-cancel tool work when the user says stop.

CURRENT PLATFORM EVIDENCE

OpenAI Realtime currently supports VAD, interruption/truncation, realtime tools and short-lived client secrets for mobile clients. Gemini Live documents persistent streamed sessions, session resumption and context compression. These are architecture references; the exact provider will be selected after

implementation audit and validation.

Target interaction

```
USER SPEAKS -> VAD -> REALTIME SESSION -> AGENT -> TOOL(S) -> VERIFIED RESULT -> STREAMING TTS
                ^                  |
                |----- USER INTERRUPTS --|
```

06 - Reference Architecture

The system is intentionally split into transport, session, orchestration, tools and domain services.

Layer	Responsibility
Android Voice UI	Mic, transcript, state indicators, stop/interruption, action confirmation, deep links
Realtime Voice Session	Streaming audio, VAD, turn state, transport, reconnect and resumption
Voice Core	Audio focus, lifecycle, cancellation, device routing, state machine, session persistence
Agent Orchestrator	Intent, context retrieval, plan/execution loop, specialist routing, response grounding
Tool Registry	Declared capabilities with input schema, permissions, confirmation, timeout, side effects
Security Gate	Authenticated identity, ownership, authorization, policy checks, tool-argument validation
Personal Tools	Placement OS domain reads and supported writes
World Tools	Live jobs, company, role, market, technology and learning retrieval
Domain Services	Existing FastAPI services, business rules, AI boundaries and data access
Storage / External Providers	Database, approved job providers, AI/voice providers, public sources

Provider abstraction

- RealtimeTransportProvider - full-duplex audio/session transport.
- SpeechToTextProvider - when using a chained voice pipeline or fallback.
- TextToSpeechProvider - spoken output provider.
- AgentModelProvider - model reasoning/orchestration layer.
- WorldSearchProvider / JobProvider - external career intelligence sources.

The business logic must not depend on one vendor. Provider adapters can change while Placement OS tools and authorization remain stable.

07 - Universal Capability and Tool System

The agent should be able to use essentially every meaningful Placement OS capability, but only through explicit, validated tools.

Tool property	Rule
Name	Stable machine-readable capability name.
Purpose	Precise description of what the tool does.
Read/write class	Clearly distinguished.
Inputs	Strict schema with validation outside the model.
Ownership	Backend derives the current user; no client-selected owner ID.
Confirmation	Explicitly declares whether user confirmation is required.
Side effects	Declared so the orchestrator can plan safely.
Timeout/cancellation	Defined for every tool.
Idempotency	Uses the existing operation-claim or business guard where needed.
Result	Structured authoritative result used for the spoken response.

Read tools

- Profile and career state
- Resume and ATS
- Skills, projects and evidence
- Career Twin / Path / Strategy / Goal
- Jobs and matches
- Applications and notes
- Interviews and assessments
- Weaknesses / missions / NBA / progress
- Professional presence / GitHub

Action tools

- Save/update job state
- Create or edit application notes
- Update supported application status
- Generate preparation plans
- Start supported interview/simulation flows
- Create reminders
- Update supported goals/preferences
- Navigate to supported app surfaces

Capability truthfulness

The agent must never claim that the app performed an action the app cannot actually perform. If a user asks for unsupported automation, the agent explains the boundary and offers the closest supported action.

08 - Action Safety: Plan, Execute, Verify

Voice control is powerful only when actions are deterministic, user-scoped and observable.

Action class	Examples	Default handling
Read-only	Show readiness, list jobs, explain a weakness, open history.	Immediate.
Reversible / moderate	Save job, add note, generate preparation, create reminder.	Validate; confirmation only where defined.
Consequential	Delete data, change major status, submit something externally if supported.	Explicit confirmation, preferably visible UI confirmation too.

Execution state machine

```
REQUESTED -> AUTHORIZED -> EXECUTING -> SUCCEEDED
                        |-> FAILED
                        |-> CANCELLED
```

Multi-step action

1. Understand the request and resolve obvious references from current context.
2. Create a bounded plan with only the required tools.
3. Validate each tool call outside the model.
4. Execute one dependency at a time unless two steps are safely parallel.
5. Verify authoritative results after each side effect.
6. Stop or reconcile if an intermediate step fails.
7. Tell the user exactly what succeeded, what did not, and what remains.

EXISTING HARDENING TO REUSE

Do not create a second mutation framework for voice. Reuse Placement OS idempotency, concurrency protection, ownership checks, AI retry budgets, AI transaction boundaries, durable operation recovery and error classification.

09 - Context, Memory and Personalization

The agent should feel continuous without retaining everything indiscriminately.

Memory layer	What belongs here	Retention rule
Turn context	Current question, referenced job, application or interview.	Ephemeral unless promoted.

Memory layer	What belongs here	Retention rule
Session state	Current mode, pending tool, current plan, active object, transcript window.	Until session ends or is safely resumed.
Conversation summary	Important facts from a long session.	Summarized and bounded.
User-approved long-term memory	Target role, preferences, explicit coaching preferences.	User-visible, editable, deletable.
Placement OS data	Actual career records and outcomes.	Controlled by existing domain ownership and retention rules.

Context retrieval

Never dump the entire Placement OS into every model request. Resolve the user intent first, retrieve the minimum relevant personal/world data, and preserve only the context needed to answer the next turn.

Current-screen awareness

The Android client may pass safe, ephemeral context such as the current screen, current job ID, application ID, interview ID or selected item. This context must not automatically become permanent memory.

10 - Career Reasoning Engine

This is where personal data and world intelligence become personalized career decisions.

Example: "Am I ready for this job?"

Evidence stream	What the agent compares
Job	Responsibilities, required skills, technologies, location, seniority.
Career Twin	Current evidence, target role alignment, readiness/evidence signals.
Resume / ATS	Keywords, gaps, evidence strength, resume risks.
Skills	Owned skills, skill gaps, demonstrated evidence.
Projects / GitHub	Proof that the user can apply the skills.
Interview history	Observed strengths, weak answers and evaluation patterns.
Preparation	What is already completed or outstanding.
Market	How relevant the role/technology is in the current target market.

Evidence language

- **KNOWN.** Directly supported by current stored or retrieved evidence.

- **INFERRED.** A reasoned conclusion based on evidence; clearly labeled as a recommendation or inference.
- **UNKNOWN.** Insufficient evidence; the agent should say so instead of hallucinating.

11 - Live Jobs, Market and Technology Intelligence

The outside-world layer should support real-time questions about the employment market.

User asks	Agent flow
"What jobs are available for me right now?"	Retrieve live jobs -> normalize/dedupe -> match against user -> rank -> explain best matches.
"Which companies are hiring freshers?"	Search current sources -> filter experience level -> geography -> return dated results.
"What is booming in AI?"	Retrieve current market/skill signals -> compare sources -> explain trend with evidence dates.
"Should I learn Spark or Kubernetes?"	Compare role relevance -> the user current skills -> market demand -> learning effort -> recommend based on target.
"What should I prepare for this role?"	Combine current role requirements -> personal gaps -> interview topics -> preparation plan.
"What is hot in Dubai?"	Use UAE-specific sources -> role/skill demand -> employer signals -> avoid importing India/US assumptions.

Job data quality rules

- Deduplicate the same opening across multiple providers before counting jobs.
- Prefer official company sources when available for current opening existence.
- Record provenance for every job and market signal.
- Do not claim a job is still open solely from stale cached data.
- Do not scrape or bypass restricted sites; use approved/public sources and adapters.
- Keep provider-specific schemas behind a normalized internal model.

12 - Preparation, Learning and Interview Coaching

The agent should convert world requirements and personal gaps into an actionable preparation sequence.

Step	Agent capability
Target role	Understand desired role, seniority, geography and job/company.
Current state	Read the user's verified skills, resume, projects, interviews and progress.
Gap analysis	Separate missing, weak, and already-proven skills.
Priority	Rank gaps by job relevance, weakness severity and time available.

Step	Agent capability
Learning path	Choose resources and practice in the order that maximizes role readiness.
Project proof	Recommend practical projects/evidence that close the gap.
Interview practice	Generate or launch targeted interview practice using the existing interview engine.
Feedback loop	Use interview/assessment outcomes to update future priorities.

Voice interview mode

The existing voice interview system becomes one specialist mode of the shared Voice Core. It should reuse the existing interview engine, evaluation logic, answer replay hardening and authenticated domain services rather than creating a parallel interview product.

13 - Security and Trust Model

The agent combines private data, untrusted external content and side-effecting tools. Security must therefore be architectural.

Threat	Required defense
Cross-user access	Backend derives identity from authenticated session; every tool enforces ownership.
Prompt injection in job/JD/GitHub content	Treat external content as untrusted data; it never becomes agent instruction.
Tool abuse	Strict schemas, authorization and policy checks outside the model.
Secret disclosure	Never expose OAuth tokens, passwords, API keys, signing keys or database credentials to the model.
Duplicate mutation	Reuse existing idempotency/concurrency guards.
False success	Verify authoritative result before speaking success.
Dangerous action	Explicit confirmation and visible confirmation where appropriate.
Sensitive audio retention	Prefer transient processing; retain only what product functionality genuinely needs.
Provider compromise	Provider adapters receive only the minimum data needed for the current request.

Mobile realtime credentialing

If a provider uses client-side realtime connections, the mobile app must receive a short-lived session credential minted by our authenticated backend rather than embedding a permanent provider API key. Current OpenAI Realtime documentation explicitly supports short-lived client secrets for mobile clients.

14 - Android Engineering Requirements

The current Android voice subsystem is useful foundation code, but the shared Voice Core must address its documented gaps.

Concern	Required design
Microphone permission	Clear disclosure + runtime permission + re-check after resume.
Audio focus	Request and respond to focus changes; do not silently continue through interruptions.
Lifecycle/configuration	Move session/recording state beyond a composition-only object; preserve or safely cancel active work.
Process death	Persist only the minimum session/operation state needed to reconcile server truth.
Background rules	Core voice mode starts from a user-visible context; do not assume unrestricted always-listening behavior.
Android 16+	Respect current foreground-service/job/runtime restrictions and test target API behavior.
Android 17+	Keep background audio assumptions conservative; user-visible sessions remain the primary model.
Audio routing	Handle speaker, wired and Bluetooth changes.
Accessibility	Keep text transcript and accessible controls available alongside voice.

CURRENT REPOSITORY FOUNDATION

The existing repository has VoiceInterviewScreen, VoiceInterviewViewModel, VoiceInterviewManager, AudioRecorder, AudioPlayer, interview audio APIs and backend faster-whisper/pyttsx3 plumbing. The existing readiness checklist also documents audio-focus and configuration/process-death gaps plus development-only voice quality limitations. These should be resolved by the new shared Voice Core rather than copied.

15 - Provider Strategy and Cost Discipline

Keep the architecture provider-neutral until VA-0 proves the correct production option.

Layer	Development path	Production decision rule
Realtime transport	Provider abstraction; local/test harness where possible.	Select after latency, Android compatibility, session behavior and cost validation.
STT	Existing faster-whisper path for dev/testing.	Replace or augment with a production-grade provider if measured quality/latency requires it.
TTS	Existing pyttsx3 path for dev/testing.	Production TTS must be platform-portable and natural enough for users; current Windows SAPI5 coupling is not sufficient alone.
Agent model	Existing AI provider architecture.	Use a provider with tool calling, streaming and acceptable reliability/cost; keep adapter boundary.
World intelligence	Existing job provider abstraction + approved public sources.	Prefer public/authorized sources with provenance, rate limits and licensing clarity.

₹0-first rule

- Do not add paid infrastructure merely to make the demo work.
- Use free/local/dev options during architecture and correctness work where practical.
- Do not embed paid provider credentials in Android.
- Do not make product availability depend on one temporary free trial.
- Measure token/audio/search costs before any production provider decision.

16 - Reliability, Recovery and Failure Handling

A premium agent is defined as much by failure behavior as by successful demos.

Failure	Expected behavior
Network drops while listening	Preserve safe session state; reconnect or fall back without corrupting the conversation.
Network drops during tool	Reconcile authoritative operation state before retrying any mutation.
AI provider times out	Cancel/stop safely; no duplicate tool or AI side effect.
TTS fails	Keep transcript/text answer available; allow retry.
STT uncertain	Show transcript; allow user correction and retry.
User says stop	Cancel audio and safe-to-cancel model/tool work; return to stable state.
Session expires	Renew/reconnect within security rules without exposing long-

Failure	Expected behavior
	lived credentials.
Process killed	Resume or reconcile from durable state; never blindly start duplicate operations.
Provider returns malformed data	Validate, reject safely, log structured error, never hallucinate a result.
World source stale	Refresh or clearly label stale information.

Graceful degradation

Realtime voice -> push-to-talk -> text agent -> normal app UI

Voice availability must never become a single point of failure for the Placement OS.

17 - Premium User Experience

The visual experience should make realtime state, trust and control obvious.

UI element	Behavior
Listening indicator	Clearly shows the microphone is active.
Live transcript	Shows what the agent heard; supports correction when important.
Thinking / tool status	Shows meaningful states such as "Checking your jobs" rather than generic indefinite loading.
Speaking indicator	Shows that the agent is producing audio.
Stop button	Always available during active agent speech/listening.
Action preview	Shows important pending changes before confirmation.
Source cards	Lets the user inspect live-world evidence while the spoken response continues.
Current context	Shows which job/application/interview the agent is acting on when relevant.
Error recovery	Offers a direct next step instead of a dead-end error screen.

Voice personality

The user can choose a presentation style such as professional, friendly, concise, detailed or strict interviewer. Personality may change tone and wording, but never changes authorization, truthfulness or safety rules.

18 - Proactive Career Coach

The agent can eventually move from reactive Q&A to relevant career guidance while respecting user consent and Android background rules.

Scenario	Example behavior
Daily briefing	"You have two application follow-ups today and your highest-priority skill gap is SQL optimization."
New job match	"I found three new jobs that closely match your target role."
Market shift	"More current listings in your target role now mention GenAI skills."
Interview readiness	"Your interview is tomorrow; the remaining high-priority preparation item is Spark optimization."
Learning reminder	"You started the SQL plan but still have the window-functions module outstanding."

CONSTRAINT

Proactive behavior should be driven by scheduled/event data and user settings, not by an always-listening microphone. Android currently restricts background microphone/foreground-service patterns, so the product should prefer notifications and user-visible voice sessions.

19 - Testing and Quality Gate

The Voice Agent gets the same engineering discipline as the existing Placement OS hardening work, extended to realtime-specific failure modes.

Test family	Examples
Unit	State machine, tool schemas, memory rules, source freshness, provider adapters.
Integration	Agent -> tool -> domain service -> authoritative result.
Realtime	VAD, turn ending, interruption, false interruption, cancellation, reconnect.
Security	Cross-account isolation, prompt injection, tool authorization, secret non-disclosure.
Mutation	Delete guards, bypass authorization, collapse action distinctions, remove verification; tests must fail.
Concurrency	Duplicate voice commands, simultaneous actions, reconnect during tool execution.

Test family	Examples
Lifecycle	Rotation, background/foreground, process death, permission revoke, audio device switch.
World intelligence	Stale data, duplicate jobs, provider disagreement, geographic mismatch, bad source injection.
Real device	Noise, Bluetooth, wired headset, calls, alarms, speaker routing, battery restrictions.
Performance	Time to speech start, first transcript, first spoken byte, tool latency, reconnect time.

Quality exit criteria

- No known P0/P1 defects in the critical Voice Agent surface.
- Critical authorization and ownership tests pass.
- Deliberate security/action mutations are caught.
- No duplicate side effects under repeated or concurrent commands.
- Interruption and cancellation remain state-correct.
- Real-world device validation passes on the agreed matrix.
- World facts expose source/freshness metadata where applicable.
- No false claims of completed actions.
- Voice can degrade to text/UI without breaking the rest of the app.

20 - Implementation Roadmap

Build in controlled phases. Do not implement everything in one large change.

Phase	Name	Scope
VA-0	Deep repository and dependency audit	Map Android/backend voice, audio, networking, auth, AI, job providers, existing interview system, navigation, tests and lifecycle constraints. Read-only.
VA-1	Shared Voice Core	Lifecycle-safe audio engine, state machine, audio focus, permission, cancellation, device routing and temporary audio handling.
VA-2	Realtime Session	Persistent realtime connection, streaming audio, VAD/turn detection, interruption, reconnect and session resumption.
VA-3	STT/TTS Abstraction	Provider interfaces, current dev path, production candidate adapters, transcript correction and latency measurement.
VA-4	Personal Tool Registry	Authenticated, user-scoped Placement OS read tools and current-screen context.

Phase	Name	Scope
VA-5	World Intelligence	Live jobs, companies, roles, skills, technology, market and learning retrieval with provenance/freshness.
VA-6	Agent Intelligence	Intent, retrieval, planning, grounding, evidence language, specialist routing and memory.
VA-7	Safe Actions	Application/job/note/preparation/ reminder/navigation actions with confirmation, idempotency and verification.
VA-8	Voice Career Coach	Natural cross-feature coaching, personalized recommendations and proactive intelligence where permitted.
VA-9	Voice Interview Coach	Shared Voice Core integrated with the existing interview engine and evaluation system.
VA-10	Security and Reliability Hardening	Prompt injection, tool abuse, action replay, concurrency, cancellation, process death, session recovery and provider failures.
VA-11	Real Device Validation	Physical-device matrix across Android versions, noise, Bluetooth, headset, calls, permissions and network conditions.
VA-12	Final Voice Agent Audit	Fresh adversarial audit; only after passing does the Voice Agent become feature-complete.

DO NOT MIX DEPLOYMENT

Release signing, production backend deployment, production URL, App Check enforcement and Play Store publishing remain outside these feature phases and are intentionally held until the desired feature set is finished.

21 - Example End-to-End Conversations

These examples define expected product behavior, not hard-coded commands.

Scenario	User	Expected agent behavior
Career status	User: "How ready am I for a Data Engineer role?"	Agent retrieves Career Twin + skills + resume/ATS + projects + interview evidence and, if needed, current role

Scenario	User	Expected agent behavior
		requirements, then explains strengths, gaps and next actions.
Live jobs	User: "Find me the best Data Engineer jobs right now."	Agent performs live job retrieval, deduplicates results, matches them to the user, ranks them and provides current source-backed recommendations.
Market question	User: "What is booming in technology right now that fits me?"	Agent combines current technology/market signals with the user's proven skills and target roles; separates facts from recommendations.
Application action	User: "Add a note to this application saying recruiter asked me to follow up Friday."	Agent uses current-screen/application context, previews the note if appropriate, executes the existing application-note action and verifies creation.
Preparation	User: "Prepare me for tomorrow's interview."	Agent identifies the interview/job, retrieves current requirements, user weaknesses and prior performance, then launches or generates the existing preparation workflow.
Multi-step	User: "Find the best matching job, check my gaps, create a plan and start an interview."	Agent plans the workflow, performs live retrieval, verifies each step, asks confirmation where needed, and hands off to interview mode.
Unsupported action	User: "Apply to every job automatically."	Agent states what the app can and cannot do; it must not claim unsupported auto-application capability.

22 - Privacy, Data Boundaries and User Control

The Voice Agent touches microphone data, career records and external sources, so privacy must be visible in the product.

Data	Default policy
Raw audio	Prefer transient processing; delete when no longer required for the requested operation.
Transcript	Retain only where required for the feature or user history; user controls should exist where practical.
Conversation memory	Only governed/appropriate memory; avoid automatic permanent storage of everything.
Career data	Use only for the app purpose and within authenticated ownership boundaries.

Data	Default policy
External search	Use minimum necessary data; show source/provenance where practical.
Secrets	Never expose to agent or voice provider unless strictly required and protected; never expose signing keys/passwords.

- Voice permission has a clear user-facing explanation before capture.
- Privacy settings can explain voice history, memory and external-data usage.
- Account deletion removes associated user data according to the existing account-deletion design.
- Logs avoid raw secrets and unnecessary sensitive content.

CURRENT POLICY NOTE

Google Play currently treats microphone and other personal/sensitive data as policy-sensitive and requires transparency, appropriate consent, secure handling, and account/data deletion mechanisms. Background collection that is not reasonably expected requires prominent disclosure and consent.

23 - Existing Repository Starting Point

What is already present in the current codebase and what the new feature should reuse.

Area	Existing foundation observed	Voice Agent implication
Android architecture	Separate API interfaces, repositories and domain repositories for major Placement OS features.	Voice tools can call existing repositories/services instead of duplicating business logic.
Voice interview	VoiceInterviewScreen/ViewModel/ Manager + AudioRecorder/AudioPlayer.	Promote reusable audio/session code into Voice Core; keep interview as a mode.
Interview backend	Interview API/service, audio answer route, evaluation and simulation infrastructure.	Voice Interview Agent can reuse existing interview engine.
AI safety	Input/output guards, retry budget, AI transaction boundary.	Voice/world tools must pass through the same guard model.
Mutation safety	Operation claims/idempotency and concurrency hardening.	Voice actions should reuse these protections.
Career intelligence	Career Twin, Career Path, Strategy, Goals, Weaknesses, NBA, Mission and Progress.	These become the agent's personal career context.
Job hunt	Provider adapters, normalization, deduplication, matching, provenance and usage controls.	Extend rather than create a parallel job-discovery stack.
Professional presence	GitHub OAuth and evidence engine already exist.	Expose approved GitHub/professional-presence reads through tools.

EXISTING VOICE LIMITATIONS TO RESOLVE

The repository's voice-readiness checklist documents two concrete Android gaps - audio focus and configuration/process-death robustness - plus development-only STT/TTS limitations. The shared Voice Core must address these before the new agent is considered production-grade.

24 - Current Research Basis

External references are used only to validate platform/architecture direction; the actual product contract remains specific to AI Placement OS.

Source	Why it matters	Link
OpenAI Realtime	Current Realtime API supports speech-to-speech sessions, VAD, interruption/truncation, tools and short-lived client secrets for browser/mobile clients.	https://developers.openai.com/api/docs/guides/realtime
OpenAI Realtime tools	Function tools are designed for application-owned business logic, approvals and private-system access.	https://developers.openai.com/api/docs/guides/realtime-mcp
OpenAI client secrets	Short-lived Realtime client secrets are intended to avoid exposing the main API key to mobile clients.	https://developers.openai.com/api/reference/resources/realtime/subresources/client_secrets/methods/create
Gemini Live session management	Live sessions stream continuously and document connection limits, session resumption and context window compression.	https://ai.google.dev/gemini-api/docs/live-api/session-management
LiveKit Agents turns	Current voice-agent guidance covers VAD, interruptions, false interruptions, endpointing and preemptive generation.	https://docs.livekit.io/agents/logic/turns/
Android background FGS restrictions	Android restricts background starts for foreground services and microphone while-in-use behavior.	https://developer.android.com/develop/background-work/services/fgs/restrictions-bg-start
Android foreground-service changes	Current Android documentation covers API 34-36 foreground service requirements and newer platform restrictions.	https://developer.android.com/develop/background-work/services/fgs/changes
Android 17 background audio	Android 17 adds background audio interaction hardening; visible activity/appropriate foreground-service context is required.	https://developer.android.com/about/versions/17/changes/bg-audio

Source	Why it matters	Link
Google Play User Data	Current Play policy treats microphone and other personal/sensitive data as sensitive and requires transparent, secure handling.	https://support.google.com/googleplay/android-developer/answer/10144311
Google Play prominent disclosure	Current guidance requires prominent disclosure and affirmative consent when sensitive access may not be reasonably expected.	https://support.google.com/googleplay/android-developer/answer/11150561

Research date: 7 October 2026. External platform behavior, pricing and policies can change; re-verify before implementation decisions that depend on current provider terms.

25 - Final Product Contract

The concise contract that future implementation prompts should preserve.

8. The Voice Agent is the universal conversational interface to the Placement OS.
9. It is realtime: streaming input/output, natural turn taking, barge-in, cancellation and recovery are core requirements.
10. It can use the authenticated user's complete Placement OS career context through controlled tools.
11. It can access current external career information through live world-intelligence tools.
12. It can operate supported app capabilities, not arbitrary phone functions.
13. The LLM never becomes the security boundary; authorization remains in the backend/tool layer.
14. Read access and action access are distinct. Consequential actions require explicit confirmation.
15. All important actions follow plan -> execute -> verify semantics.
16. Existing idempotency, concurrency, AI safety and ownership hardening is reused.
17. World data carries provenance and freshness so current claims can be audited.
18. Memory is bounded, purposeful and user-controllable.
19. The system degrades gracefully to push-to-talk/text/UI when realtime voice is unavailable.
20. The existing voice interview feature becomes one specialist mode of the shared Voice Core.
21. Deployment/publishing is intentionally not part of the Voice Agent feature phases.

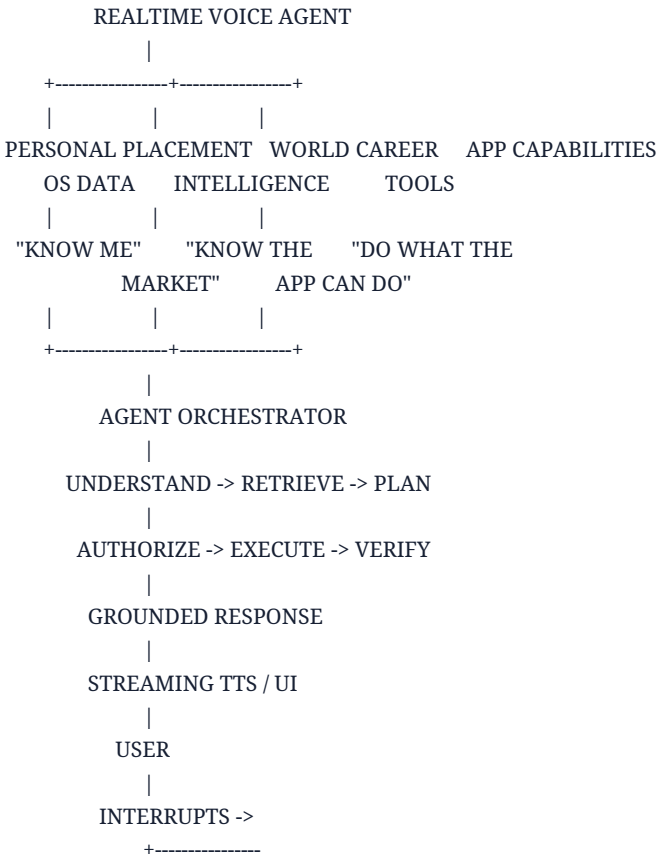
DEFINITION OF DONE

The Voice Agent is feature-complete only after VA-12 confirms the critical realtime, world-intelligence, user-data, app-control, security, lifecycle, concurrency, privacy and real-device invariants. No known P0/P1 issue may remain.

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One-Page Mental Model

The entire system in one view.



FINAL VISION

"Know me. Know the market. Control my Placement OS. Talk to me in realtime."